

WASP-50b

R-0708

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_241201 · created 2026-07-27T22:31:51+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460645.74 +/- 0.12 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.049
Transit depth (Rp/Rs)^2	3.0 +/- 1.68 [%]
Orbital Inclination (inc)	89.04 +/- 2.84
Airmass coefficient 1 (a1)	1.97 +/- 1.06
Airmass coefficient 2 (a2)	-0.47 +/- 0.65
Scatter in the residuals of the lightcurve fit is	52.8 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	3.75
Transit Duration (day)	0.04 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.049 vs 0.1404 (deviation 0.67σ)
Duration fit vs published	0.04 d vs 0.0754 d (deviation 0.96σ)
Mid-time O–C	-35.1 min
Depth SNR	3.5
Residual scatter	52.8 %

Light curve

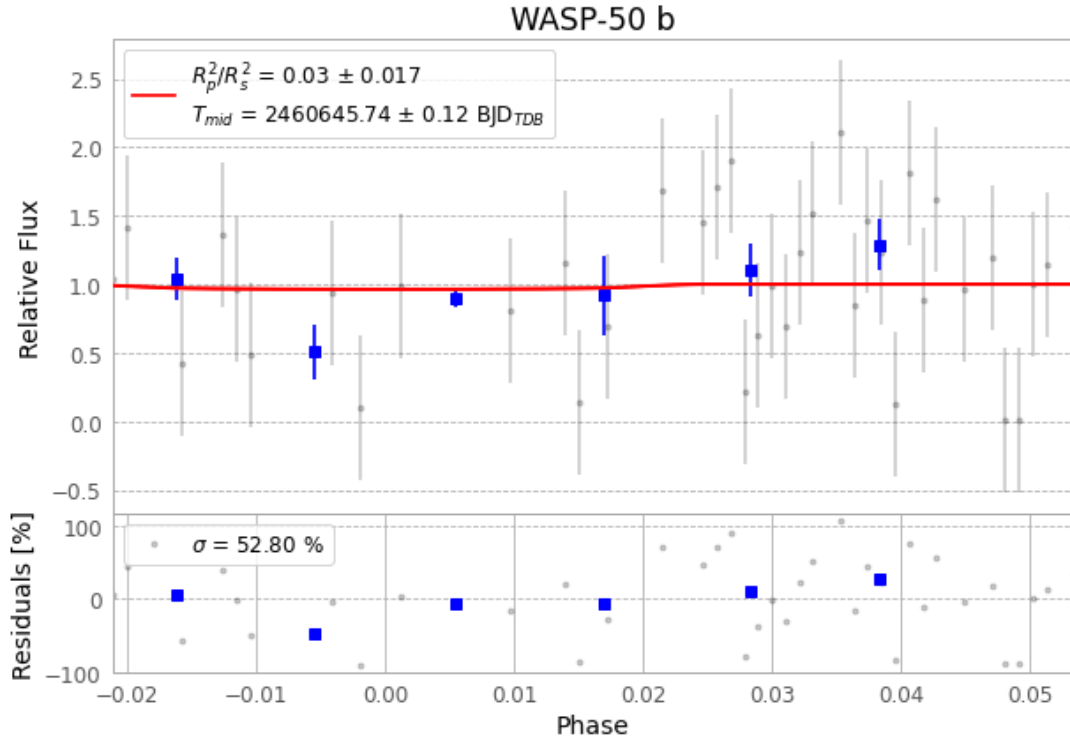


Figure 1 — FinalLightCurve_WASP-50 b_2024-11-30.png (SHA-256 c50af0ce8be893bc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [334, 421], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a89a8cbf8b535746...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

71 FITS frames (47 MB), WASP-50241201043915.FITS → WASP-50241201080915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-241201.log (ecdadfce8bff...), FinalLightCurve_WASP-50 b_2024-11-30.png (c50af0ce8be8...), FinalLightCurve_WASP-50 b_2024-11-30.csv (5903dcbcd366...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 22:31Z.